



Total Quality Management & Six Sigma

Six Sigma DMAIC Project: Reducing Defects in Wooden Toy Manufacturing

Submitted By: - Learning Team-5

Submitted To: -

Dr. Akansha Jain

Riya Gupta 25PGDM039

Dev Bhatia 25PGDM073

Kaushal Goyal 25PGDM084

Jatin Kumar Rajput 25PGDM199

1. EXECUTIVE SUMMARY

Project Overview

This Six Sigma DMAIC project focuses on reducing manufacturing defects at Chandan Wooden Toys Pvt. Ltd. The project uses the 28-day production and quality dataset in the Excel workbook to understand current performance across Carving/Shaping, Paint/Finish and Joint/Assembly.

Problem

During the 28-day baseline, 10,909 toys were produced and 593 were recorded as defective. The resulting defect rate was 5.44%, with a yield of 94.56%.

Baseline Performance

The baseline defect rate is 5.44%. Using the three defect opportunities per unit specified in the workbook, the DPMO is 18,119.6. The project target is to reduce the overall defect rate toward $\leq 2.50\%$ while maintaining production volume.

Key Findings

- Paint/Finish is the largest defect category, with 277 defects (46.7% of total recorded defects).
- Joint/Assembly is the second priority, with 178 defects (30.0%).
- Carving/Shaping accounts for 138 defects (23.3%).
- Paint/Finish and Joint/Assembly together account for 76.7% of the recorded defects.
- The P-chart identifies three special-cause points: Days 8, 15 and 21.
- The Pareto analysis identifies where to focus; it does not by itself prove causation.

Improvement Direction

The workbook proposes standardising paint application, using visual defect standards, defining minimum drying time, introducing first-piece approval, improving joint positioning with a fixture or poka-yoke, checking tools daily, and using a common defect-definition guide.

Control

The proposed Control Plan uses daily KPI and P-chart monitoring, category-wise tracking, first-piece approval for every batch and weekly SOP compliance checks. Green, yellow and red reaction logic is used to guide routine operation, supervisor review and corrective action.

KPI Snapshot

KPI	Baseline
Total Production	10,909
Total Defective	593
Defect Rate	5.44%
Yield	94.56%
DPMO	18,119.6
Paint/Finish	277 (46.7%)
Joint/Assembly	178 (30.0%)
Carving/Shaping	138 (23.3%)
P-chart special-cause points	3 — Days 8, 15 and 21
Project Target	$\leq 2.50\%$ defect rate

2. PROJECT CHARTER

Item	Project Information
Project Title	Reduction of manufacturing defects in wooden toy production
Business Problem	During the 28-day baseline, 593 defective toys were recorded from 10,909 produced, resulting in a 5.44% defect rate.
Goal Statement	Reduce the overall defect rate from 5.44% toward a project target of $\leq 2.50\%$ while maintaining production volume.
CTQ	Defect-free wooden toy with acceptable carving/shaping, paint/finish and joint/assembly quality.
Scope – In	Carving/Shaping, Paint/Finish, Joint/Assembly, inspection, defect recording and rework/rejection.
Scope – Out	Marketing, pricing, distribution and customer-service activities outside manufacturing quality.
Stakeholders	Production Supervisor, Quality Inspector, Operators and Operations Manager.

The project is focused on a measurable manufacturing CTQ and uses the supplied baseline dataset as the starting point for the Measure and subsequent DMAIC phases.

2.1 PROCESS OVERVIEW – SIPOC

Supplier	Input	Process	Output	Customer
Wood/material suppliers	Wood components	Receive materials	Finished wooden toy	End customer
Paint/adhesive suppliers	Paint / adhesive	Carve / shape	Accepted product	Retail / distribution
Tool / fixture suppliers	Tools / fixtures	Paint / finish	Defect records	Internal operations
Production support	Work instructions	Assemble joints	Rework / rejected output	Quality team
		Inspect & record		

Process Flow: Receive materials → Carve / shape → Paint / finish → Assemble joints → Inspect & record

3. DEFINE PHASE

The Define phase establishes the business problem, project goal, CTQ, scope and process covered by the project. The project is focused on manufacturing quality and uses the 28-day baseline dataset supplied in the workbook.

The main quality problem is manufacturing defects across Carving/Shaping, Paint/Finish and Joint/Assembly, with an overall baseline defect rate of 5.44%. The project target is $\leq 2.50\%$ while maintaining production volume.

4. MEASURE PHASE – DATA & PERFORMANCE ANALYSIS

4.1 Baseline KPI Analysis

KPI	Basis	Result	Meaning
Total Production	Sum of 28 daily observations	10,909	Baseline volume
Total Defective	Sum of recorded defectives	593	Baseline quality loss
Defect Rate	Defectives / production	5.44%	Baseline defect performance
Yield	1 – defect rate	94.56%	Good output rate
DPMO	Defectives/(units × 3 opportunities) × 1,000,000	18,119.6	Normalised defect metric
P-chart special-cause points	Variable-sample 3σ limits	3	Days 8, 15, 21

4.2 Calculation Details

Defect Rate = Total Defective / Total Production = 593 / 10,909 = 0.05435879 = 5.44%

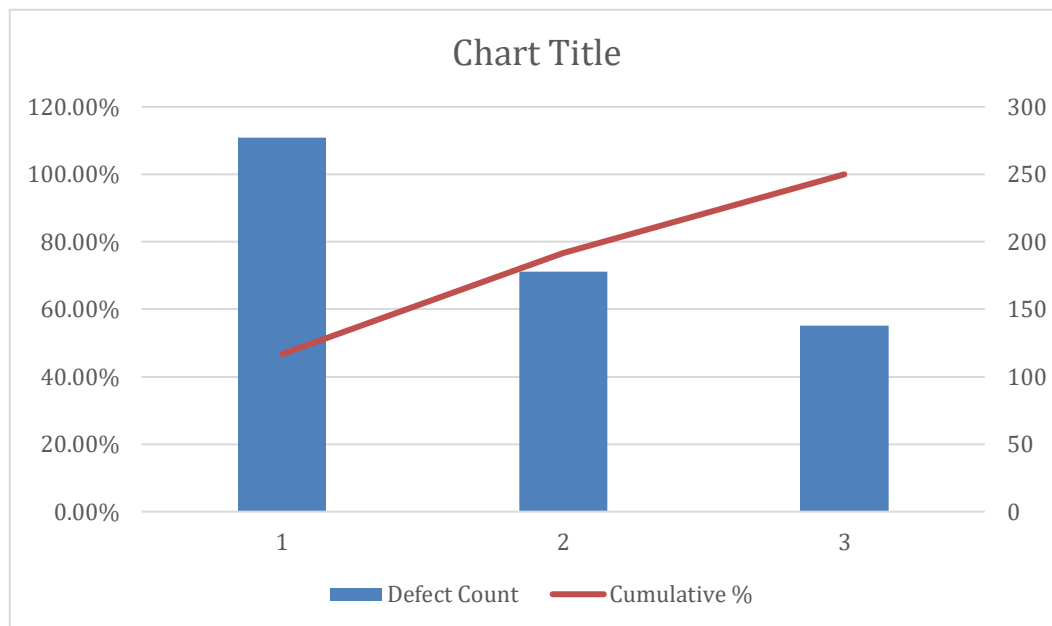
Yield = 1 – Defect Rate = 1 – 0.05435879 = 0.94564121 = 94.56%

$DPMO = \text{Total Defective} / (\text{Total Production} \times 3 \text{ opportunities}) \times 1,000,000 = 593 / (10,909 \times 3) \times 1,000,000 = 18,119.6$

$P\text{-bar} = \text{Total Defective} / \text{Total Production} = 593 / 10,909 = 0.05435879$

4.3 Pareto Analysis

Rank	Defect Category	Defect Count	Cumulative	Cumulative %
1	Paint/Finish	277	277	46.7%
2	Joint/Assembly	178	455	76.7%
3	Carving/Shaping	138	593	100.0%



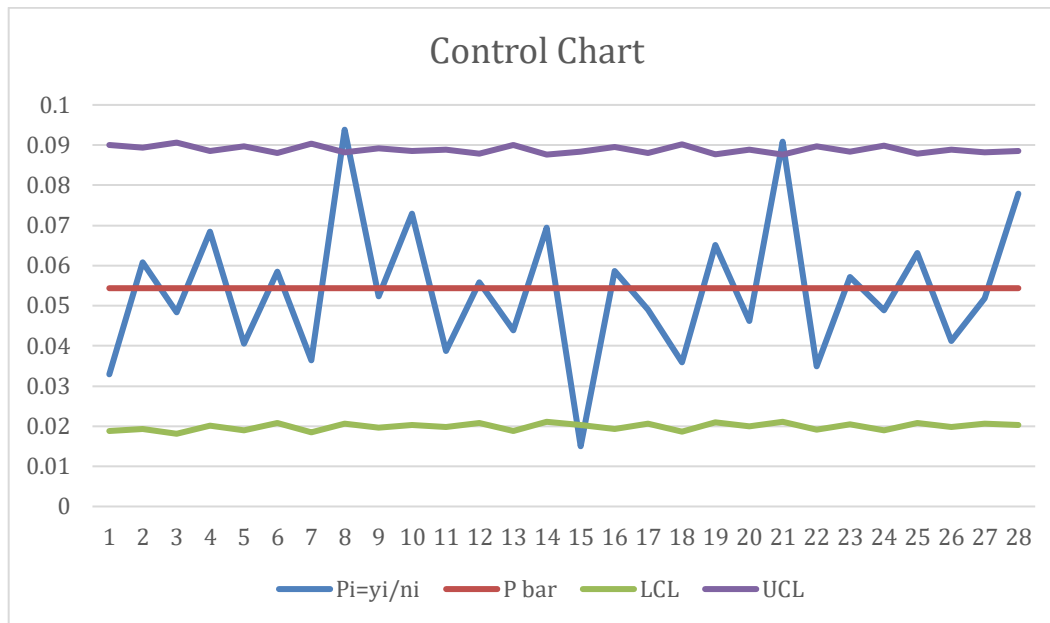
The chart above is the Pareto chart from the Excel analysis. Paint/Finish is the largest contributor, followed by Joint/Assembly and Carving/Shaping. The first two categories account for 76.7% of the recorded defects.

4.4 P-Chart Analysis

The P-chart in the Excel uses the daily proportion of defective toys ($P_i = y_i/n_i$), the overall P-bar, and variable-sample 3σ limits. The control limits change with the daily production quantity.

The Excel analysis identifies special-cause points on Days 8, 15 and 21.

Excel P-chart — Days 1–14



Excel P-chart — Days 15–28

4.5 P-Chart Calculation Table

Day	Ni	yi	$P_i = y_i/n_i$	$\bar{P}$	S_p	LCL	UCL
1	365	12	0.03287671	0.05435879	0.01186729	0.01875691	0.08996066
2	378	23	0.06084656	0.05435879	0.01166144	0.01937447	0.08934311
3	352	17	0.04829545	0.05435879	0.01208445	0.01810545	0.09061212
4	395	27	0.06835443	0.05435879	0.01140774	0.02013557	0.08858200
5	370	15	0.04054054	0.05435879	0.01178684	0.01899828	0.08971929
6	410	24	0.05853659	0.05435879	0.01119712	0.02076744	0.08795013
7	358	13	0.03631285	0.05435879	0.01198275	0.01841053	0.09030704
8	405	38	0.09382716	0.05435879	0.01126602	0.02056072	0.08815685
9	382	20	0.05235602	0.05435879	0.01160022	0.01955811	0.08915946
10	398	29	0.07286432	0.05435879	0.01136466	0.02026480	0.08845277
11	388	15	0.03865979	0.05435879	0.01151018	0.01982824	0.08888933
12	412	23	0.05582524	0.05435879	0.01116990	0.02084907	0.08786850
13	365	16	0.04383562	0.05435879	0.01186729	0.01875691	0.08996066
14	418	29	0.06937799	0.05435879	0.01108945	0.02109044	0.08762713
15	400	6	0.01500000	0.05435879	0.01133622	0.02035014	0.08836743
16	375	22	0.05866667	0.05435879	0.01170799	0.01923481	0.08948276
17	408	20	0.04901961	0.05435879	0.01122453	0.02068521	0.08803236
18	362	13	0.03591160	0.05435879	0.01191636	0.01860969	0.09010788
19	415	27	0.06506024	0.05435879	0.01112946	0.02097041	0.08774716
20	390	18	0.04615385	0.05435879	0.01148063	0.01991689	0.08880068
21	418	38	0.09090909	0.05435879	0.01108945	0.02109044	0.08762713
22	372	13	0.03494624	0.05435879	0.01175511	0.01909346	0.08962411
23	402	23	0.05721393	0.05435879	0.01130798	0.02043485	0.08828273
24	368	18	0.04891304	0.05435879	0.01181882	0.01890232	0.08981525
25	412	26	0.06310680	0.05435879	0.01116990	0.02084907	0.08786850
26	388	16	0.04123711	0.05435879	0.01151018	0.01982824	0.08888933
27	405	21	0.05185185	0.05435879	0.01126602	0.02056072	0.08815685
28	398	31	0.07788945	0.05435879	0.01136466	0.02026480	0.08845277

5. ANALYZE PHASE – ROOT CAUSE ANALYSIS

5.1 Pareto Interpretation

Paint/Finish is the largest defect category with 277 defects (46.7% of total recorded defects). Joint/Assembly is the second priority. The Pareto identifies where to focus; it does not by itself prove causation.

5.2 Fishbone Analysis

Cause Category	Causes Listed in the Excel
MACHINE	Tool wear; Paint equipment variation; Maintenance / calibration
MAN	Operator technique; Training / experience; Fatigue / handover
MATERIAL	Wood condition / moisture; Paint viscosity / batch; Adhesive consistency
METHOD	Paint quantity / coats; Drying time; Assembly sequence; First-piece approval
MEASUREMENT	Defect classification; Inspection consistency; Recording accuracy
ENVIRONMENT	Dust / cleanliness; Temperature / humidity; Workstation condition

Effect: HIGH MANUFACTURING DEFECT RATE.

Leading root-cause hypothesis from the Excel: variation in standard work and process controls—especially paint application/drying and visual acceptance criteria—may be contributing to Paint/Finish defects. This must be validated with operator, material, method, equipment and environment data during the improvement pilot.

5.3 5 Why – Paint / Finish

Why 1	Why 2	Why 3	Why 4	Root-cause hypothesis
Finish varies between units	Application / drying conditions vary	Method may depend on operator judgement	Key parameters are not consistently documented or checked	Insufficient standard work and visual controls
Defects are detected after production	Abnormal conditions are not caught early	No first-piece gate is evidenced in baseline	Checks occur after production	Introduce first-piece approval and process checks

6. IMPROVE PHASE – IMPROVEMENT RECOMMENDATIONS

Priority	Problem / Cause	Improvement	Expected Benefit	Effort	Decision
1	Paint application variation	Standard Paint/Finish SOP + visual defect standard	Reduce operator variation	Low	Pilot
2	Drying variation	Minimum drying time + first-piece approval	Catch abnormal finish early	Low	Pilot
3	Joint positioning variation	Assembly fixture / poka-yoke	Improve repeatability	Medium	Pilot
4	Tool wear	Daily tool check + replacement trigger	Reduce carving defects	Low	Pilot
5	Defect classification variation	Common defect-definition guide	Improve measurement consistency	Low	Implement

6.1 FMEA Risk Assessment

Failure Mode	Effect	S	Potential Cause	O	Current Detection	D	RPN	Recommended Action	Priority
Incorrect paint application	Poor finish / rejection	8	Inconsistent method	7	Visual inspection	5	280	Standard SOP + visual standard	High
Insufficient drying	Finish damage / rejection	7	Variable drying time	6	Final inspection	6	252	Minimum drying time + first-piece check	High
Wrong joint positioning	Weak / incorrect assembly	8	Manual positioning variation	5	Final inspection	5	200	Fixture / poka-yoke	Medium
Tool wear	Poor carving / shaping	6	Worn tool	5	Operator inspection	6	180	Tool check + replacement trigger	Medium
Incorrect defect classification	Poor data quality	5	Unclear criteria	5	Supervisor review	7	175	Defect classification guide	Lower

6.2 Pilot Evaluation

Run the selected controls on a pilot basis and compare the post-improvement defect rate with the 5.44% baseline. Track total defects and each defect category daily. Do not claim the 2.50% target as achieved until post-improvement data confirms it.

7. RISK ASSESSMENT

Failure Mode / Risk	RPN	Priority	Recommended Action
Incorrect paint application	280	High	Standard SOP + visual standard
Insufficient drying	252	High	Minimum drying time + first-piece check
Wrong joint positioning	200	Medium	Fixture / poka-yoke
Tool wear	180	Medium	Tool check + replacement trigger
Incorrect defect classification	175	Lower	Defect classification guide

The FMEA values above are taken from the Excel Improve sheet. The highest RPN is associated with incorrect paint application, followed by insufficient drying.

8. CONTROL PLAN

KPI / CTQ	Baseline	Target / Standard	Frequency	Owner	Control Method	Trigger	Reaction	Record	Status
Overall defect rate	5.44%	≤2.50% project target	Daily	Quality	P-chart + KPI	Above target / special cause	Investigate, correct and verify	Daily quality log	Ready
Paint/Finish defects	277	Reduce from baseline	Daily	Production	Category trend	Trend increases	Check SOP, application, drying	Quality log	Ready
Joint/Assembly defects	178	Reduce from baseline	Daily	Assembly	Category trend	Trend increases	Check fixture / method	Quality log	Ready
Carving/Shaping defects	138	Reduce from baseline	Daily	Production	Category trend	Trend increases	Check tools / material	Quality log	Ready
First-piece approval	Not available	100% of batches	Every batch	Quality	Checklist	Approval missing	Hold batch until approved	Batch checklist	Implement
SOP compliance	Not available	≥95%	Weekly	Supervisor	Audit checklist	<95%	Retrain / correct standard work	Audit record	Implement

8.1 Control Reaction Logic

GREEN: process remains within target/control limits; continue standard work.

YELLOW: deterioration or near-limit performance; supervisor review.

RED: target missed or special-cause signal; perform root-cause investigation, corrective action and effectiveness verification.

9. PROJECT CLOSURE & CONCLUSION

DMAIC Phase	Key Outcome
DEFINE	Manufacturing defects were defined as the quality problem, with a 5.44% baseline defect rate.
MEASURE	The 28-day dataset contains 10,909 toys produced and 593 defective toys, giving 18,119.6 DPMO.
ANALYZE	Paint/Finish is the largest defect category. The Pareto shows where to focus, while the Fishbone and 5 Why provide root-cause hypotheses that require validation.
IMPROVE	The workbook proposes standard work, drying control, first-piece approval, assembly repeatability, tool checks and consistent defect classification.

CONTROL

Daily KPI/P-chart monitoring, first-piece approval and weekly SOP audits are proposed to sustain improvement.

The baseline analysis shows a clear opportunity to reduce manufacturing defects, with Paint/Finish as the main area for improvement attention. The recommended actions are directly linked to the improvement options and root-cause hypotheses in the Excel workbook. The $\leq 2.50\%$ figure remains a project target and should only be reported as achieved after post-improvement data confirms it.

10. PEER EVALUATION

Complete the following table with the actual team members and their contributions before submission.

Team member	Primary contribution	Details
Kaushal Goyal	Excel-based Data & Performance Analysis (Measure/Analyze phase)	Led the Excel-based analysis of the project dataset, including data organization, defect calculations, baseline metrics, Pareto analysis and P-chart analysis. Supported the Measure and Analyze sections with accurate calculations and interpretation.
Jatin Kumar Rajput	Excel-based DMAIC Analysis (Improve/Control phase)	Worked on the Excel-based Improve and Control sections, including improvement options, FMEA/risk assessment, control plan and checking the consistency of calculations and project outputs.
Riya Gupta	Report & Presentation Development	Contributed to structuring and formatting the written report and presentation slides. Supported the preparation of key findings, improvement recommendations and presentation content.
Dev Bhatia	Report & Presentation Development	Contributed to preparing and organizing the presentation, summarizing the DMAIC findings and supporting the development of the written report, recommendations and final presentation flow.

11. AI USAGE DECLARATION – AI COLLABORATION

AI Tools Used

ChatGPT, Gemini, Claude AI

Purpose of AI Use

- Brainstorming and structuring the DMAIC project
- Supporting Excel calculations and analysis
- Suggesting possible root causes and improvements
- Supporting report formatting and presentation

Team Validation

AI suggestions were reviewed and validated by the team before being included in the final project.

APPENDIX A – 28-DAY BASELINE DATA

Day	Toys Produced	Total Defective	Carving/Shaping	Paint/Finish	Joint/Assembly	Defect Rate	Yield Rate
1	365	12	3	5	4	3.29%	96.71%
2	378	23	6	9	8	6.08%	93.92%
3	352	17	5	8	4	4.83%	95.17%
4	395	27	6	12	9	6.84%	93.16%
5	370	15	3	7	5	4.05%	95.95%
6	410	24	5	12	7	5.85%	94.15%
7	358	13	3	6	4	3.63%	96.37%
8	405	38	9	18	11	9.38%	90.62%
9	382	20	5	8	7	5.24%	94.76%
10	398	29	7	14	8	7.29%	92.71%
11	388	15	3	7	5	3.87%	96.13%
12	412	23	6	10	7	5.58%	94.42%
13	365	16	4	7	5	4.38%	95.62%
14	418	29	8	13	8	6.94%	93.06%
15	400	6	1	3	2	1.50%	98.50%
16	375	22	4	12	6	5.87%	94.13%
17	408	20	4	11	5	4.90%	95.10%
18	362	13	3	7	3	3.59%	96.41%
19	415	27	6	13	8	6.51%	93.49%
20	390	18	4	8	6	4.62%	95.38%
21	418	38	8	17	13	9.09%	90.91%
22	372	13	3	6	4	3.49%	96.51%
23	402	23	6	10	7	5.72%	94.28%
24	368	18	4	9	5	4.89%	95.11%
25	412	26	6	12	8	6.31%	93.69%
26	388	16	4	8	4	4.12%	95.88%
27	405	21	5	9	7	5.19%	94.81%
28	398	31	7	16	8	7.79%	92.21%

APPENDIX B – CALCULATION SUMMARY

Calculation	Excel Basis	Result
Defect Rate	Defectives / production	593 / 10,909 = 5.44%
Yield	1 – defect rate	94.56%
DPMO	Defectives/(units × 3 opportunities) × 1,000,000	18,119.6
P-bar	Total defective / total production	0.05435879
Special-cause points	Variable-sample 3σ limits	Days 8, 15, 21